

Real Analysis Mth125

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August 30, 2026

1 Schedule

The reference book for mth125 is Principle of Mathematical Analysis written by Walter Rudin. There are 11 chapters in this book. They are

- ch1: The real and complex number systems
- ch2: Basic Topology
- ch3: Numerical sequences and series
- ch4: Continuity
- ch5: Differentiation
- ch6: The Riemann-Stieltjes integral
- ch7: Sequences and series of functions
- ch8: some special functions
- ch9: Functions of several variables
- ch10: Integration of Differential forms
- ch11: The Lebesgue theory

We can decompose these 11 chapters into six topics:

- Topic 1(ch1+ch2): Origin of numbers, and algebraic and geometric traits of real number system.

- Topic 2(ch3+ch4): Switching the order of mapping and taking limit
- Topic 3(ch5+ch6): Calculus of single variable functions
- Topic 4(ch7+ch8): Special functions and function space
- Topic 5(ch9+ch10): Calculus of multi-variable functions
- Topic 6(ch11): Modern Integration

In each semester, we have 13 teaching weeks, one of which is reserved for final review. The teaching schedule is as follows:

- week 1: Topic 1(ch1+ch2)
- Week 2-3: Topic 2(ch3+ch4)
- Week 4-6: Topic 3(ch5+ch6)
- Week 7-8: Topic 4(ch7+ch8)
- Week 9-11: Topic 5(ch9+ch10)
- Week 12: Topic 6(ch11)
- Week 13: Final review.

There is only one in-semester class test. Test 1 should be arranged in Wednesday 7:00 p.m.–8:30 p.m. at week 7, covering topic 1-3. Topic 6 is an advanced topic, which will not be covered in final exam.

2 Syllabus

Mth125 is challenge for our students. We should encourage our students to understand real analysis under the assistance of AI tools. There is a huge bank of exercise problems. We should encourage them to work on these problems. All assessments problems are selected from this bank.

This task seems to be an impossible mission. Therefore, we should change our teaching philosophy. We don't teach theorem proofs step by step in our lectures. We should behave like eagles, demonstrating the origins and connections among different concepts. In tutorials, we should demonstrate how to use AI to solve problems in

the test banks. Most importantly, we should demonstrate how to review the solution of AI line by line. Hopefully, we can cultivate the critical thinking ability of our students in the era of AI. After this, we should encourage our students to try all the problems in the test bank. We should make full use of our office hours to help students to build up their critical thinking ability.

We will demonstrate the implementation demos in the following sections.

2.1 Topic 1: Origin of numbers, and algebraic and geometric properties of real number system.

We should complete this topic in one week. We have 3 hours lecture and 1 hour tutorial. So we need to introduce all the concepts about real number system and their connections in 3 hours. Then we will use 1 hour tutorial to demonstrate how to review the AI output, and how to modify the proof representation. We should not talk about too many problems in 1 hour tutorial, possibly just one problem.

Implementation Demo:

3 hours lecture:

- Introduce the history of numbers: rational numbers come from counting and measurement. Irrational numbers exist, but it must be interpreted as limit of rational numbers.
- Explain what we mean by limits of a sequence. $\epsilon - N$ language should be explained geometrically.
- Introduce the extension from rational numbers to real number systems.
 - Algebraic properties of rational numbers.
 - real numbers will inherit the algebraic properties of rational numbers.
 - Introduce completeness of real number systems through Cauchy principle. Explain why Cauchy principle is easier to use than definition of limits because we don't need to know the limit point in advance.
 - compactness criteria of real number system and various techniques of showing the existence of limits.

1 hour tutorial:

- We will choose one exercise problem on how to show existence of limit.

- We use AI tools to get the solutions first. Based on these AI solutions, we conduct review process, explaining which part is reasonable, and which is non-sense.
- We should carefully explain the reviewing process, and make sure the students realize that AI can make mistakes, we must be responsible for its mistakes if we choose to reply on its solution.

Problem 1: If $p_k > 0$, where $k = 1, 2, \dots$, and

$$\lim_{n \rightarrow \infty} \frac{p_n}{p_1 + p_2 + \dots + p_n} = 0, \quad \lim_{n \rightarrow \infty} a_n = a,$$

then show that

$$\lim_{n \rightarrow \infty} \frac{p_1 a_n + p_2 a_{n-1} + \dots + p_n a_1}{p_1 + p_2 + \dots + p_n} = a.$$

Solution from AI (DeepSeek):

Proof. Let

$$P_n = \sum_{k=1}^n p_k, \quad T_n = \frac{\sum_{k=1}^n p_k a_{n-k+1}}{P_n}.$$

Since $a_n \rightarrow a$, the sequence $\{a_n\}$ is bounded (**why it is bounded, and what do we mean by "bounded"?**). Thus there exists $M > 0$ such that

$$|a_n| \leq M$$

for all n (**This is what we mean by "bounded"**).

Given $\varepsilon > 0$, since $a_n \rightarrow a$, there exists N_1 such that for all $k > N_1$,

$$|a_k - a| < \frac{\varepsilon}{2}.$$

(**This is just the definition of limit, $\epsilon - N$ language**)

Also, since

$$\frac{p_n}{P_n} \rightarrow 0,$$

there exists N_2 such that for all $n > N_2$,

$$\frac{p_n}{P_n} < \frac{\varepsilon}{4M}.$$

(This is also just the definition of limit, $\epsilon - N$ language, but there is a small error for $\frac{p_n}{P_n} < \frac{\varepsilon}{4M}$, we are supposed to have

$$\frac{p_k}{P_k} < \frac{\varepsilon}{4MN_2}, \quad \forall n - N_2 + 1 \leq k \leq n.$$

)

Let $N = \max\{N_1, N_2\}$ (What's going on? why do you start to define such an N ? Furthermore, we have

$$\frac{p_k}{P_k} < \frac{\varepsilon}{4MN_2}, \quad \forall n - N_2 + 1 \leq n - N + 1 \leq k \leq n.$$

). For $n > 2N$, we split the sum defining T_n into two parts:

$$T_n - a = \frac{\sum_{k=1}^n p_k(a_{n-k+1} - a)}{P_n}.$$

Write

$$T_n - a = \frac{1}{P_n} \sum_{k=1}^{n-N} p_k(a_{n-k+1} - a) + \frac{1}{P_n} \sum_{k=n-N+1}^n p_k(a_{n-k+1} - a).$$

For the first sum, if $k \leq n - N$, then

$$n - k + 1 \geq N + 1 > N_1,$$

so

$$|a_{n-k+1} - a| < \frac{\varepsilon}{2}.$$

Therefore,

$$\left| \frac{1}{P_n} \sum_{k=1}^{n-N} p_k(a_{n-k+1} - a) \right| \leq \frac{\varepsilon}{2} \cdot \frac{\sum_{k=1}^{n-N} p_k}{P_n} \leq \frac{\varepsilon}{2}.$$

For the second sum, since it contains only N terms,

$$\left| \frac{1}{P_n} \sum_{k=n-N+1}^n p_k(a_{n-k+1} - a) \right| \leq \frac{1}{P_n} \sum_{k=n-N+1}^n p_k \cdot 2M.$$

Because $p_k > 0$ and $p_k/P_k \rightarrow 0$, for sufficiently large n and for $k \geq n - N + 1$ we have

$$\frac{p_k}{P_n} \leq \frac{p_k}{P_k} < \frac{\varepsilon}{4MN}.$$

(This is wrong! We should have, for $n - N \leq k \leq n$,

$$\frac{p_k}{P_n} \leq \frac{\varepsilon}{4MN_2}, \quad \forall n \geq 2N = 2\max\{N_1, N_2\} \geq N_2$$

)

Thus

$$\frac{1}{P_n} \sum_{k=n-N+1}^n p_k \cdot 2M \leq 2M \sum_{k=n-N+1}^n \frac{p_k}{P_n} \leq 2M \sum_{k=n-N+1}^n \frac{\varepsilon}{4M} = \frac{\varepsilon}{2}.$$

(Come on! Why do we need this whole paragraph? What's for? By the way, this is wrong! We should have

$$\frac{1}{P_n} \sum_{k=n-N+1}^n p_k \cdot 2M \leq 2M \sum_{k=n-N+1}^n \frac{p_k}{P_n} \leq 2M \sum_{k=n-N+1}^n \frac{\varepsilon}{4M} = \frac{\varepsilon}{2} \frac{N}{N_2}.$$

)

Combining the two estimates gives

$$|T_n - a| < \varepsilon$$

for all sufficiently large n . (I get it! You are trying to verify the definition of convergence, i.e. for any $\epsilon > 0$, we need to verify that

$$|T_n - a| < \epsilon, \forall n \geq N$$

where N should be specifically determined. We should focus on this single problem. In order to guarantee $|T_n - a|$ could be as small as possible, it is decomposed into two parts in AI's solution:

$$T_n - a = \frac{1}{P_n} \sum_{k=1}^{n-N} p_k (a_{n-k+1} - a) + \frac{1}{P_n} \sum_{k=n-N+1}^n p_k (a_{n-k+1} - a).$$

For part I: $(a_{n-k+1} - a)$, we know $N + 1 \leq n - k + 1 \leq n$. So $|a_{n-k+1} - a| \leq \frac{\epsilon}{2}$ can be small for $n - k + 1$ is large enough. So

$$|\frac{1}{P_n} \sum_{k=1}^{n-N} p_k (a_{n-k+1} - a)| \leq \frac{\epsilon}{2} \frac{1}{P_n} \sum_{k=1}^{n-N} p_k \leq \frac{\epsilon}{2}$$

For part II: $\frac{1}{P_n} \sum_{k=n-N+1}^n p_k(a_{n-k+1} - a)$, there are actually only finite terms for $n - N + 1 \leq k \leq n$, i.e. there are only N terms. Then $(a_{n-k+1} - a)$ is of course bounded by some constant M , and

$$\left| \frac{1}{P_n} \sum_{k=n-N+1}^n p_k(a_{n-k+1} - a) \right| \leq M \sum_{k=n-N+1}^n \frac{p_k}{P_n} \leq M \sum_{k=n-N+1}^n \frac{p_k}{P_k}$$

because $\frac{p_k}{P_k} \rightarrow 0$ for $k \rightarrow \infty$, so these N terms can be as small as possible. Then we can make sure $\frac{p_k}{P_n} \leq \frac{\epsilon}{2MN}$, so

$$\left| \frac{1}{P_n} \sum_{k=n-N+1}^n p_k(a_{n-k+1} - a) \right| \leq M \sum_{k=n-N+1}^n \frac{p_k}{P_k} \leq MN \frac{\epsilon}{2N_2M} = \frac{\epsilon}{2} \frac{N}{N_2}.$$

In total, we have

$$|T_n - a| < \epsilon \left(\frac{1}{2} + \frac{N}{2N_2} \right) \neq \epsilon.$$

It seems that we have not verified the definition of convergence:

$$\forall \epsilon > 0, \exists N > 0, s.t. |T_n - a| < \epsilon$$

There must be something wrong! How can we figure it out? In fact, everything dates back to the decomposition of $T_n - a$. It should be

$$T_n - a = \frac{1}{P_n} \sum_{k=1}^{n-N_2} p_k(a_{n-k+1} - a) + \frac{1}{P_n} \sum_{k=n-N_2+1}^n p_k(a_{n-k+1} - a).$$

In order to guarantee $\frac{1}{P_n} \sum_{k=1}^{n-N_2} p_k(a_{n-k+1} - a) < \frac{\epsilon}{2}$, we have to make sure

$$n - k + 1 \geq n - (n - N_2) + 1 = N_2 + 1 \geq N_1$$

So for $\frac{p_n}{P_n} \rightarrow 0$, we have to choose $N_2 \geq N_1 - 1$, so that

$$\left| \frac{p_n}{P_n} \right| \leq \frac{\epsilon}{4MN_2}$$

Then choose $N = \max\{N_1, N_2\}$, when $n \geq N$, we have

$$|T_n - a| = \frac{1}{P_n} \sum_{k=1}^{n-N_2} p_k |a_{n-k+1} - a| + \frac{1}{P_n} \sum_{k=n-N_2+1}^n p_k |a_{n-k+1} - a|$$

$$\begin{aligned}
&\leq \frac{\epsilon}{2} \frac{1}{P_n} \sum_{k=1}^{n-N_2} p_k + 2M \sum_{k=n-N_2+1}^n \frac{p_k}{P_n} \\
&\leq \frac{\epsilon}{2} + 2M \sum_{k=n-N_2+1}^n \frac{p_k}{P_k} \\
&\leq \frac{\epsilon}{2} + 2M \sum_{k=n-N_2+1}^n \frac{\epsilon}{4MN_2} \\
&\leq \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.
\end{aligned}$$

Now you will understand why do we need to choose $N = \max\{N_1, N_2\}$, because we want to make sure both parts can be as small as possible.)

Hence

$$\lim_{n \rightarrow \infty} T_n = a,$$

which is exactly

$$\lim_{n \rightarrow \infty} \frac{p_1 a_n + p_2 a_{n-1} + \cdots + p_n a_1}{p_1 + p_2 + \cdots + p_n} = a.$$

□

(Conclusion: This solution is not completely acceptable! There are some loopholes in its logic!

- You will realize that the key part is the decomposition of $T_n - a$. Understanding how to make sure two parts as small as possible is the key idea of analysis.
- When you try to prove the convergence, you need to start with the decomposition of $T_n - a$. You should try your best to figure out how to guarantee both parts could be as small as possible. Once you figure out this, then you almost understand how to prove it.
- To write down a proof, we need to follow the definition. There is a common pattern for you to follow.
 - First, write down for $\epsilon > 0$,
 - then, explain how do we specify an integer N .
 - explain why $|T_n - a| < \epsilon$.
 - Last, write down conclusion that $T_n \rightarrow a$ as $n \rightarrow \infty$.

- You will realize that the order of analysing part is different from the written proof. Please get used to this as soon as possible. This is your first step in real analysis!

)
Modified Solution (Human):

Proof. Let

$$P_n = \sum_{k=1}^n p_k, \quad T_n = \frac{\sum_{k=1}^n p_k a_{n-k+1}}{P_n}.$$

Since $a_n \rightarrow a$, the sequence $\{a_n\}$ is bounded (This is a theorem, you can refer to it without proof!). Thus there exists $M > 0$ such that

$$|a_n| \leq M$$

for all n .

Given $\varepsilon > 0$, since $a_n \rightarrow a$, there exists N_1 such that for all $k > N_1$,

$$|a_k - a| < \frac{\varepsilon}{2}.$$

(This is the definition of limit, you can literally copy it down, but here you just particularly choose $\frac{\varepsilon}{2}$ and N_1 in your definition!)

Also, since

$$\frac{p_n}{P_n} \rightarrow 0,$$

there exists $N_2 \geq N_1 - 1$ such that for all $n > N_2$, (Since our written proof is in an opposite order of analysis. It may be strange too you at first, but when you reach the end of the proof you will understand why do we choose it in such way!)

$$\frac{p_n}{P_n} < \frac{\varepsilon}{4MN_2}.$$

Let $N = \max\{N_1, N_2\}$. (Again, It may be strange too you at first, but when you reach the end of the proof you will understand we have to choose such N to guarantee both parts are small enough!)

For $n > 2N$, we split the sum defining T_n into two parts:

$$T_n - a = \frac{\sum_{k=1}^n p_k (a_{n-k+1} - a)}{P_n}.$$

Write

$$T_n - a = \frac{1}{P_n} \sum_{k=1}^{n-N_2} p_k(a_{n-k+1} - a) + \frac{1}{P_n} \sum_{k=n-N_2+1}^n p_k(a_{n-k+1} - a).$$

For the first sum, if $k \leq n - N_2$, then

$$n - k + 1 \geq N_2 + 1 > N_1,$$

so

$$|a_{n-k+1} - a| < \frac{\varepsilon}{2}.$$

Therefore,

$$\left| \frac{1}{P_n} \sum_{k=1}^{n-N_2} p_k(a_{n-k+1} - a) \right| \leq \frac{\varepsilon}{2} \cdot \frac{\sum_{k=1}^{n-N_2} p_k}{P_n} \leq \frac{\varepsilon}{2}.$$

For the second sum, since it contains only N_2 terms,

$$\left| \frac{1}{P_n} \sum_{k=n-N_2+1}^n p_k(a_{n-k+1} - a) \right| \leq \sum_{k=n-N_2+1}^n \frac{p_k}{P_n} \cdot 2M.$$

For $k \geq n - N_2 + 1$ we have

$$\frac{p_k}{P_n} \leq \frac{p_k}{P_k} < \frac{\varepsilon}{4MN_2}.$$

Thus

$$\frac{1}{P_n} \sum_{k=n-N_2+1}^n p_k \cdot 2M \leq 2M \sum_{k=n-N_2+1}^n \frac{p_k}{P_k} \leq 2M \sum_{k=n-N_2+1}^n \frac{\varepsilon}{4MN_2} = \frac{\varepsilon}{2}.$$

Combining the two estimates gives: for $\varepsilon > 0$, there exists $N = \max\{N_1, N_2\}$ such that

$$|T_n - a| < \varepsilon$$

for all sufficiently large $n \geq N$. Hence

$$\lim_{n \rightarrow \infty} T_n = a,$$

which is exactly

$$\lim_{n \rightarrow \infty} \frac{p_1 a_n + p_2 a_{n-1} + \cdots + p_n a_1}{p_1 + p_2 + \cdots + p_n} = a.$$

□

3 Topic 2: Series and continuity

We should also complete this topic in one week. We have 3 hours lecture and 1 hour tutorial. We need to introduce all concepts of series $\sum_{n=1}^{\infty} a_n$, transformation of series $\sum_{n=1}^{\infty} f(a_n)$ for some function $f(x)$. Most importantly we have to explain continuity by explaining when do we have

$$\lim_{n \rightarrow \infty} f(a_n) = f(\lim_{n \rightarrow \infty} a_n)$$

We also have to show how to manipulate special series such as $\sum_{n=0}^{\infty} x^n$ and $\sum_{n=0}^{\infty} \frac{x^n}{n!}$. We should talk about 2 problems in tutorial, one for series and one for continuity.

Implementation Demo:

3 hours lecture:

- Introduce the definition of series $\sum_{n=1}^{\infty} f(a_n)$.
- Manipulate series through algebraic operation: addition, subtraction, multiplication and division. Use special series as an example.
- introduce the limit of a function and its continuity. Explain that continuity can guarantee that we can switch the order of limit and mapping

$$\lim_{n \rightarrow \infty} f(a_n) = f(\lim_{n \rightarrow \infty} a_n)$$

- talk about properties of continuous functions.

1 hour tutorial:

- We should choose two problems, one for series and another for continuity.
- We use AI tools to get the solutions first. Based on these AI solutions, we conduct review process, explaining which part is reasonable, and which is not acceptable.

Problem 1:

Compute $\sum_{k=1}^{\infty} \frac{k^3 + 6k^2 + 11k + 5}{(k+3)!}$.

Solution from AI (DeepSeek):

Solution: We need to compute

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{k^3 + 6k^2 + 11k + 5}{(k+3)!}.$$

Step 1: Rewrite the numerator

We look for constants A, B, C, D such that

$$k^3 + 6k^2 + 11k + 5 = A(k+3)(k+2)(k+1) + B(k+3)(k+2) + C(k+3) + D.$$

Expanding the right-hand side:

$$A(k^3 + 6k^2 + 11k + 6) + B(k^2 + 5k + 6) + C(k+3) + D.$$

Thus

$$Ak^3 + (6A + B)k^2 + (11A + 5B + C)k + (6A + 6B + 3C + D).$$

Comparing coefficients with $k^3 + 6k^2 + 11k + 5$, we get

$$A = 1, \quad B = 0, \quad C = 0, \quad D = -1.$$

Therefore,

$$k^3 + 6k^2 + 11k + 5 = (k+3)(k+2)(k+1) - 1.$$

Step 2: Express the term using factorial

Since

$$(k+3)! = (k+3)(k+2)(k+1)k!,$$

we have

$$\frac{k^3 + 6k^2 + 11k + 5}{(k+3)!} = \frac{(k+3)(k+2)(k+1) - 1}{(k+3)!}.$$

Thus

$$\frac{k^3 + 6k^2 + 11k + 5}{(k+3)!} = \frac{1}{k!} - \frac{1}{(k+3)!}.$$

Step 3: Sum the series

Therefore,

$$\sum_{k=1}^n \frac{k^3 + 6k^2 + 11k + 5}{(k+3)!} = \sum_{k=1}^n \left(\frac{1}{k!} - \frac{1}{(k+3)!} \right).$$

This telescopes:

$$\sum_{k=1}^n \frac{1}{k!} - \sum_{k=1}^n \frac{1}{(k+3)!}.$$

The second sum can be shifted:

$$\sum_{k=1}^n \frac{1}{(k+3)!} = \sum_{j=4}^{n+3} \frac{1}{j!}.$$

Hence

$$\sum_{k=1}^n \left(\frac{1}{k!} - \frac{1}{(k+3)!} \right) = \left(\frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} \right) - \left(\frac{1}{(n+1)!} + \frac{1}{(n+2)!} + \frac{1}{(n+3)!} \right).$$

Step 4: Take the limit

As $n \rightarrow \infty$,

$$\frac{1}{(n+1)!} + \frac{1}{(n+2)!} + \frac{1}{(n+3)!} \rightarrow 0.$$

Therefore,

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{k^3 + 6k^2 + 11k + 5}{(k+3)!} = \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} = 1 + \frac{1}{2} + \frac{1}{6} = \frac{5}{3}.$$

Thus,

$$\boxed{\frac{5}{3}}$$

(**Conclusion: This is completely acceptable!**)

Solution from human: We know $\sum_{k=0}^{\infty} \frac{1}{k!} = e$. So We want to know if $k^3 + 6k^2 + 11k + 5$ can be reduced by $k + 3$. To this end, we use long division.

We perform long division of

$$k^3 + 6k^2 + 11k + 5$$

by

$$k + 3.$$

Long Division Steps

$$\begin{array}{r|rrrr}
 & k^2 & +3k & +2 & \\
 k+3 & k^3 & +6k^2 & +11k & +5 \\
 & k^3 & +3k^2 & & \\
 \hline
 & & 3k^2 & +11k & \\
 & & 3k^2 & +9k & \\
 \hline
 & & & 2k & +5 \\
 & & & 2k & +6 \\
 \hline
 & & & & -1
 \end{array}$$

Thus the quotient is

$$k^2 + 3k + 2$$

and the remainder is

$$-1.$$

Therefore,

$$\frac{k^3 + 6k^2 + 11k + 5}{(k+3)!} = \frac{(k+3)(k^2 + 3k + 2)}{(k+3)!} - \frac{1}{(k+3)!} = \frac{k^2 + 3k + 2}{(k+2)!} - \frac{1}{(k+3)!}.$$

Further factoring gives

$$k^2 + 3k + 2 = (k+1)(k+2).$$

Hence

$$\begin{aligned}
 \frac{k^3 + 6k^2 + 11k + 5}{(k+3)!} &= \frac{(k+2)(k+1)}{(k+2)!} - \frac{1}{(k+3)!} \\
 &= \frac{1}{k!} - \frac{1}{(k+3)!}
 \end{aligned}$$

Then

$$\sum_{k=1}^n \left(\frac{1}{k!} - \frac{1}{(k+3)!} \right) = \left(\frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} \right) - \left(\frac{1}{(n+1)!} + \frac{1}{(n+2)!} + \frac{1}{(n+3)!} \right).$$

As $n \rightarrow \infty$,

$$\frac{1}{(n+1)!} + \frac{1}{(n+2)!} + \frac{1}{(n+3)!} \rightarrow 0.$$

Therefore,

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{k^3 + 6k^2 + 11k + 5}{(k+3)!} = \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} = 1 + \frac{1}{2} + \frac{1}{6} = \frac{5}{3}.$$

(Conclusion: Long division is a very good method for algebraic manipulation. We also need this technique in antiderivative!)

Problem 2:

Suppose functions $f(x), g(x)$ are defined in some neighborhood of 0, $g(x) > 0$, $\lim_{x \rightarrow 0} \frac{f(x)}{g(x)} = 1$; and as $n \rightarrow \infty$, $\alpha_{mn} \rightarrow 0$ ($m = 1, 2, \dots, n$). That is, $\forall \varepsilon > 0$, $\exists N(\varepsilon) > 0$ such that when $n > N(\varepsilon)$, we have

$$\max_{1 \leq m \leq n} |\alpha_{mn}| < \varepsilon,$$

Show that if $\alpha_{mn} \neq 0$, then

$$\lim_{n \rightarrow \infty} \sum_{m=1}^n f(\alpha_{mn}) = \lim_{n \rightarrow \infty} \sum_{m=1}^n g(\alpha_{mn})$$

provided the right-hand limit exists.

Solution from AI (DeepSeek):

Proof. Since

$$\lim_{x \rightarrow 0} \frac{f(x)}{g(x)} = 1,$$

for every $\varepsilon > 0$, there exists $\delta > 0$ such that if $0 < |x| < \delta$, then

$$\left| \frac{f(x)}{g(x)} - 1 \right| < \varepsilon.$$

Equivalently,

$$|f(x) - g(x)| < \varepsilon |g(x)|.$$

Because $\alpha_{mn} \rightarrow 0$ uniformly in m , for the above $\delta > 0$, there exists N such that for all $n > N$,

$$\max_{1 \leq m \leq n} |\alpha_{mn}| < \delta.$$

Since $\alpha_{mn} \neq 0$, we have $0 < |\alpha_{mn}| < \delta$ for all $m = 1, \dots, n$. Hence for $n > N$,

$$|f(\alpha_{mn}) - g(\alpha_{mn})| < \varepsilon |g(\alpha_{mn})|$$

for every $m = 1, \dots, n$.

Summing over m ,

$$\left| \sum_{m=1}^n f(\alpha_{mn}) - \sum_{m=1}^n g(\alpha_{mn}) \right| \leq \sum_{m=1}^n |f(\alpha_{mn}) - g(\alpha_{mn})| < \varepsilon \sum_{m=1}^n |g(\alpha_{mn})|.$$

Since $g(x) > 0$ near 0, for sufficiently large n , $g(\alpha_{mn}) > 0$, (This is not precise. We only know $\max_{1 \leq m \leq n} \alpha_{mn} \leq \delta$, but we don't know whether $g(x) > 0$, for $0 \leq x \leq \delta$. There could be $\delta_1 > 0$, such that $g(x) > 0$ for $0 \leq x \leq \delta_1$. So to guarantee that

$$|f(x) - g(x)| \leq \epsilon |g(x)|,$$

and

$$g(x) > 0$$

we should choose $\delta' = \min\{\delta, \delta_1\}$. Note that $\alpha_{mn} \rightarrow 0$ ($m = 1, 2, \dots, n$), then for given $\delta' > 0$, we can choose $N(\delta')$ such that

$$\max_{1 \leq m \leq n} \alpha_{mn} < \delta.$$

)

so

$$\sum_{m=1}^n |g(\alpha_{mn})| = \sum_{m=1}^n g(\alpha_{mn}).$$

Thus

$$\left| \sum_{m=1}^n f(\alpha_{mn}) - \sum_{m=1}^n g(\alpha_{mn}) \right| < \varepsilon \sum_{m=1}^n g(\alpha_{mn}).$$

Now, if

$$\lim_{n \rightarrow \infty} \sum_{m=1}^n g(\alpha_{mn})$$

exists and is finite, then the sequence of partial sums $\sum_{m=1}^n g(\alpha_{mn})$ is bounded, say by $M > 0$. Hence for $n > N$,

$$\left| \sum_{m=1}^n f(\alpha_{mn}) - \sum_{m=1}^n g(\alpha_{mn}) \right| < \varepsilon M.$$

Since $\varepsilon > 0$ is arbitrary, it follows that

$$\lim_{n \rightarrow \infty} \left(\sum_{m=1}^n f(\alpha_{mn}) - \sum_{m=1}^n g(\alpha_{mn}) \right) = 0.$$

(We could also verify the definition of limit of sequence. To this end, we just need to use the definition of limit of function. First, we show that $\sum_{m=1}^n g(\alpha_{mn})$ is bounded by M due its convergence. Then by the definition of function limit. There exists δ_1 such that

$$|f(x) - g(x)| \leq \frac{\epsilon}{M} |g(x)|$$

Because $g(x)$ is positive near 0, i.e. there exists $\delta_2 > 0$, such that

$$g(x) > 0, \forall 0 \leq x \leq \delta_2.$$

Now we choose $\delta = \min\{\delta_1, \delta_2\}$, then because $\alpha_{mn} \rightarrow 0$ ($m = 1, 2, \dots, n$), so there exists $N(\delta)$ such that

$$\alpha_{mn} < \delta$$

Then we have

$$\left| \sum_{m=1}^n f(\alpha_{mn}) - \sum_{m=1}^n g(\alpha_{mn}) \right| \leq \frac{\epsilon}{M} \sum_{m=1}^n |g(\alpha_{mn})| = \frac{\epsilon}{M} \sum_{m=1}^n g(\alpha_{mn}) \leq \frac{\epsilon}{M} M = \epsilon.$$

) Therefore, provided the limit on the right exists,

$$\lim_{n \rightarrow \infty} \sum_{m=1}^n f(\alpha_{mn}) = \lim_{n \rightarrow \infty} \sum_{m=1}^n g(\alpha_{mn}).$$

□

(Conclusion: This is only partially acceptable! There are still some loopwholes in its logic! We will try to write down the proof in two different ways.)

Proof from Human:

Proof 1: Verification of limit: we need to verify the definition of limit, i.e. for any $\epsilon > 0$, there exists N , such that

$$\left| \sum_{m=1}^n f(\alpha_{mn}) - \sum_{m=1}^n g(\alpha_{mn}) \right| \leq \epsilon.$$

Proof. Since the limit of $\sum_{m=1}^n g(\alpha_{mn})$ exists, then it is bounded by M . Because $\lim_{x \rightarrow 0} \frac{f(x)}{g(x)} = 1$, there exists δ_1 such that

$$|f(x) - g(x)| \leq \frac{\epsilon}{M} |g(x)|$$

Moreover, $g(x)$ is positive near 0, i.e. there exists $\delta_2 > 0$, such that

$$g(x) > 0, \forall 0 \leq x \leq \delta_2.$$

Now we choose $\delta = \min\{\delta_1, \delta_2\}$, then because $\alpha_{mn} \rightarrow 0$ ($m = 1, 2, \dots, n$), so there exists $N(\delta)$ ($N(\delta)$ means N depends on δ , which is uniquely determined by ϵ , thus N is uniquely determined by ϵ) such that

$$\alpha_{mn} < \delta$$

Then we have

$$\left| \sum_{m=1}^n f(\alpha_{mn}) - \sum_{m=1}^n g(\alpha_{mn}) \right| \leq \frac{\epsilon}{M} \sum_{m=1}^n |g(\alpha_{mn})| = \frac{\epsilon}{M} \sum_{m=1}^n g(\alpha_{mn}) \leq \frac{\epsilon}{M} M = \epsilon.$$

Therefore, by the definition of limit, we know $\lim_{n \rightarrow \infty} \sum_{m=1}^n f(\alpha_{mn}) - \sum_{m=1}^n g(\alpha_{mn}) = 0$, i.e.

$$\lim_{n \rightarrow \infty} \sum_{m=1}^n f(\alpha_{mn}) = \lim_{n \rightarrow \infty} \sum_{m=1}^n g(\alpha_{mn}),$$

(the right hand side exists so the equation makes sense.) □

Proof 2: We use the fact that limit exists is equivalent to upper limit and lower limit are equal.:

Since

$$|f(x) - g(x)| = \left| \frac{f(x)}{g(x)} - 1 \right| |g(x)|$$

Then

$$\begin{aligned} \left| \sum_{m=1}^n f(\alpha_{mn}) - \sum_{m=1}^n g(\alpha_{mn}) \right| &\leq \sum_{m=1}^n |f(\alpha_{mn}) - g(\alpha_{mn})| \\ &\leq \sum_{m=1}^n \left| \frac{f(\alpha_{mn})}{g(\alpha_{mn})} - 1 \right| |g(\alpha_{mn})| \leq \max_{1 \leq m \leq n} \left| \frac{f(\alpha_{mn})}{g(\alpha_{mn})} - 1 \right| \sum_{m=1}^n |g(\alpha_{mn})|. \end{aligned}$$

Then

$$0 \leq \limsup_{n \rightarrow \infty} \left| \sum_{m=1}^n f(\alpha_{mn}) - \sum_{m=1}^n g(\alpha_{mn}) \right| \leq \limsup_{n \rightarrow \infty} \max_{1 \leq m \leq n} \left| \frac{f(\alpha_{mn})}{g(\alpha_{mn})} - 1 \right| \limsup_{n \rightarrow \infty} \sum_{m=1}^n |g(\alpha_{mn})| \leq 0$$

because α_{mn} goes to 0 uniformly and $g(x)$ is positive near 0, then

$$\sum_{m=1}^n |g(\alpha_{mn})| = \sum_{m=1}^n g(\alpha_{mn})$$

which approaches to a fix limit when $n \rightarrow \infty$.

4 Topic 3: Calculus of single variable functions

We have three weeks to complete this topic. Therefore, we have 9 hours lectures to cover all the concepts of diffitiability and integrability. We will have another 3 hours tutorials to talk about various kinds of exercise problems.

Implementation Demo: 9 hours Lectures:

- Introduce the definition of derivative and criteria of existence of derivatives.
- Mean value theorem, Taylor expansion and inequality(including extreme value, inequality).
- Introduce the definition of integrability, and criteria of integrability.
- Integral inequalities and improper integrals.

3 hour Tutorials:

Problem 1: Let $p(x) = x$, $q(x) = 1 - x$, and let $f(x)$ be a polynomial such that $f(x) \geq p(x)$ and $f(x) \geq q(x)$ for all $x \in (-\infty, +\infty)$. Prove that $f(\frac{1}{2}) > \frac{1}{2}$.

Proof from AI (DeepSeek):

Proof. We are given

$$p(x) = x, \quad q(x) = 1 - x,$$

and a polynomial $f(x)$ satisfying

$$f(x) \geq p(x), \quad f(x) \geq q(x)$$

for all $x \in (-\infty, \infty)$. We want to prove

$$f\left(\frac{1}{2}\right) > \frac{1}{2}.$$

At $x = \frac{1}{2}$, we have

$$p\left(\frac{1}{2}\right) = \frac{1}{2}, \quad q\left(\frac{1}{2}\right) = \frac{1}{2}.$$

Thus the two lines $y = x$ and $y = 1 - x$ intersect at the point

$$\left(\frac{1}{2}, \frac{1}{2}\right).$$

Since $f(x)$ is a polynomial, it is continuous everywhere. The conditions

$$f(x) \geq x, \quad f(x) \geq 1 - x$$

imply in particular that

$$f\left(\frac{1}{2}\right) \geq \frac{1}{2}.$$

We claim that equality cannot occur. Suppose, for contradiction, that

$$f\left(\frac{1}{2}\right) = \frac{1}{2}.$$

Consider the polynomial

$$F(x) = f(x) - x.$$

Since $f(x) \geq x$ for all x , we have

$$F(x) \geq 0$$

for all x , and

$$F\left(\frac{1}{2}\right) = f\left(\frac{1}{2}\right) - \frac{1}{2} = 0.$$

Thus F has a global minimum at $x = \frac{1}{2}$. Therefore, as a differentiable function,

$$F'\left(\frac{1}{2}\right) = 0.$$

That is,

$$f'\left(\frac{1}{2}\right) - 1 = 0,$$

so

$$f' \left(\frac{1}{2} \right) = 1.$$

Similarly, define

$$G(x) = f(x) - (1 - x).$$

The condition $f(x) \geq 1 - x$ gives

$$G(x) \geq 0$$

for all x , and

$$G \left(\frac{1}{2} \right) = f \left(\frac{1}{2} \right) - \left(1 - \frac{1}{2} \right) = 0.$$

Thus G also has a global minimum at $x = \frac{1}{2}$, so

$$G' \left(\frac{1}{2} \right) = 0.$$

That is,

$$f' \left(\frac{1}{2} \right) + 1 = 0,$$

so

$$f' \left(\frac{1}{2} \right) = -1.$$

This yields

$$1 = -1,$$

which is impossible.

Therefore, the assumption $f(\frac{1}{2}) = \frac{1}{2}$ must be false. Since we already have

$$f \left(\frac{1}{2} \right) \geq \frac{1}{2},$$

it follows that

$$f \left(\frac{1}{2} \right) > \frac{1}{2}.$$

This completes the proof.

(Conclusion: This proof is conducted as by contradiction, which is completely acceptable.)

□

Problem 2: Let $f(x)$ have a second derivative on $[a, b]$ with $f'(a) = f'(b) = 0$. Prove that there exists $\xi \in (a, b)$ such that

$$|f''(\xi)| \geq \frac{4}{(b-a)^2} |f(b) - f(a)|.$$

Proof from AI (DeepSeek):

Proof. We prove the statement by contradiction using Taylor's theorem.

Let

$$M = \frac{4}{(b-a)^2} |f(b) - f(a)|.$$

Assume, for contradiction, that

$$|f''(x)| < M$$

for all $x \in (a, b)$.

Since f has a second derivative on $[a, b]$, we can apply Taylor's theorem with the Lagrange form of the remainder around a and b .

Step 1: Taylor expansion about a

Because $f'(a) = 0$, for any $x \in (a, b]$,

$$f(x) = f(a) + \frac{f''(\xi_1)}{2} (x-a)^2$$

for some $\xi_1 \in (a, x)$. Hence

$$|f(x) - f(a)| = \frac{|f''(\xi_1)|}{2} (x-a)^2 < \frac{M}{2} (x-a)^2.$$

In particular, taking $x = \frac{a+b}{2}$,

$$\left| f\left(\frac{a+b}{2}\right) - f(a) \right| < \frac{M}{2} \left(\frac{b-a}{2}\right)^2 = \frac{M(b-a)^2}{8}.$$

Step 2: Taylor expansion about b

Similarly, since $f'(b) = 0$, for any $x \in [a, b)$,

$$f(x) = f(b) + \frac{f''(\xi_2)}{2}(x - b)^2$$

for some $\xi_2 \in (x, b)$. Thus

$$|f(x) - f(b)| < \frac{M}{2}(x - b)^2.$$

Taking $x = \frac{a+b}{2}$,

$$\left| f\left(\frac{a+b}{2}\right) - f(b) \right| < \frac{M}{2} \left(\frac{b-a}{2}\right)^2 = \frac{M(b-a)^2}{8}.$$

Step 3: Contradiction

Using the triangle inequality,

$$|f(b) - f(a)| \leq \left| f\left(\frac{a+b}{2}\right) - f(a) \right| + \left| f\left(\frac{a+b}{2}\right) - f(b) \right|.$$

Therefore,

$$|f(b) - f(a)| < \frac{M(b-a)^2}{8} + \frac{M(b-a)^2}{8} = \frac{M(b-a)^2}{4}.$$

But

$$M = \frac{4}{(b-a)^2} |f(b) - f(a)|,$$

so

$$\frac{M(b-a)^2}{4} = |f(b) - f(a)|.$$

Hence we obtain

$$|f(b) - f(a)| < |f(b) - f(a)|,$$

which is impossible.

Thus our assumption was false, and there must exist some $\xi \in (a, b)$ such that

$$|f''(\xi)| \geq \frac{4}{(b-a)^2} |f(b) - f(a)|.$$

This completes the proof.

Conclusion: This proof is completely acceptable!

□

Problem 3: Prove that the set $A \equiv \left\{ \alpha \mid \forall x > 0, \left(1 + \frac{1}{x}\right)^{x+\alpha} > e \right\}$ has a minimum, and find this minimum.

Proof from AI (DeepSeek):

Proof. We are given the set

$$A = \left\{ \alpha \mid \forall x > 0, \left(1 + \frac{1}{x}\right)^{x+\alpha} > e \right\}.$$

We need to prove that A has a minimum and find it.

Step 1: Rewrite the inequality

Taking natural logarithms, the inequality

$$\left(1 + \frac{1}{x}\right)^{x+\alpha} > e$$

is equivalent to

$$(x + \alpha) \ln \left(1 + \frac{1}{x}\right) > 1.$$

Thus

$$\alpha \ln \left(1 + \frac{1}{x}\right) > 1 - x \ln \left(1 + \frac{1}{x}\right).$$

Since $\ln(1 + \frac{1}{x}) > 0$ for all $x > 0$, we can divide to obtain

$$\alpha > \frac{1 - x \ln \left(1 + \frac{1}{x}\right)}{\ln \left(1 + \frac{1}{x}\right)}.$$

Define

$$\varphi(x) = \frac{1 - x \ln \left(1 + \frac{1}{x}\right)}{\ln \left(1 + \frac{1}{x}\right)}.$$

Then the defining condition for A is

$$\alpha > \varphi(x)$$

for all $x > 0$. Therefore,

$$A = \left\{ \alpha \mid \alpha > \sup_{x>0} \varphi(x) \right\}.$$

Since A consists of all real numbers greater than some fixed number, it will have a minimum if and only if the supremum of φ is attained, that is, if φ has a maximum on $(0, \infty)$. Then the minimum of A will be that maximum value.

Step 2: Simplify $\varphi(x)$

Let

$$t = \frac{1}{x} > 0.$$

Then

$$\ln\left(1 + \frac{1}{x}\right) = \ln(1 + t),$$

and

$$x \ln\left(1 + \frac{1}{x}\right) = \frac{\ln(1 + t)}{t}.$$

Thus

$$\varphi(x) = \frac{1 - \frac{\ln(1+t)}{t}}{\ln(1+t)} = \frac{t - \ln(1+t)}{t \ln(1+t)}.$$

So we define

$$\psi(t) = \frac{t - \ln(1+t)}{t \ln(1+t)}, \quad t > 0.$$

We need to find

$$\sup_{t>0} \psi(t).$$

Step 3: Study the function $\psi(t)$

First, note that as $t \rightarrow 0^+$,

$$\ln(1+t) = t - \frac{t^2}{2} + O(t^3),$$

so

$$t - \ln(1+t) = \frac{t^2}{2} + O(t^3),$$

and

$$t \ln(1+t) = t^2 + O(t^3).$$

Therefore,

$$\psi(t) \rightarrow \frac{1}{2}$$

as $t \rightarrow 0^+$.

As $t \rightarrow \infty$,

$$\ln(1+t) \sim \ln t,$$

so

$$t - \ln(1+t) \sim t,$$

and

$$t \ln(1+t) \sim t \ln t.$$

Hence

$$\psi(t) \sim \frac{1}{\ln t} \rightarrow 0.$$

Thus $\psi(t) \rightarrow 0$ as $t \rightarrow \infty$.

Since $\psi(t)$ is continuous and positive on $(0, \infty)$, and tends to 0 at infinity while tending to $1/2$ at 0, it is natural to conjecture that the maximum is $1/2$, attained in the limit $t \rightarrow 0^+$. But $t = 0$ is not in the domain. We must check whether $\psi(t) < 1/2$ for all $t > 0$.

Consider

$$\frac{1}{2} - \psi(t) = \frac{1}{2} - \frac{t - \ln(1+t)}{t \ln(1+t)} = \frac{t \ln(1+t) - 2t + 2 \ln(1+t)}{2t \ln(1+t)}.$$

Let

$$h(t) = t \ln(1+t) - 2t + 2 \ln(1+t) = (t+2) \ln(1+t) - 2t.$$

We claim $h(t) < 0$ for all $t > 0$, which would imply $\psi(t) < 1/2$.

Step 4: Show $h(t) < 0$

We have

$$h'(t) = \ln(1+t) + \frac{t+2}{1+t} - 2 = \ln(1+t) + \frac{t+2-2-2t}{1+t} = \ln(1+t) - \frac{t}{1+t}.$$

For $t > 0$, it is well known that

$$\ln(1+t) > \frac{t}{1+t}$$

for $t > 0$. Indeed, the function

$$r(t) = \ln(1+t) - \frac{t}{1+t}$$

has derivative

$$r'(t) = \frac{1}{1+t} - \frac{1}{(1+t)^2} = \frac{t}{(1+t)^2} > 0,$$

and $r(0) = 0$, so $r(t) > 0$. Therefore,

$$h'(t) > 0$$

for $t > 0$. Since $h(0) = 0$, it follows that $h(t) > 0$ for $t > 0$.

Thus

$$\frac{1}{2} - \psi(t) > 0$$

for all $t > 0$, so

$$\psi(t) < \frac{1}{2}$$

for all $t > 0$.

Step 5: Supremum is $1/2$

We have shown

$$\sup_{t>0} \psi(t) = \frac{1}{2},$$

and the supremum is not attained in the domain $t > 0$. Therefore,

$$\sup_{x>0} \varphi(x) = \frac{1}{2}.$$

Hence

$$A = \left\{ \alpha \mid \alpha > \frac{1}{2} \right\} = \left(\frac{1}{2}, \infty \right).$$

Step 6: Minimum of A

The set $A = (1/2, \infty)$ does **not** contain its greatest lower bound $1/2$. Therefore, A has **no minimum**.

Conclusion

The set A does not have a minimum. Its infimum is

$$\inf A = \frac{1}{2}.$$

Thus the statement “ A has a minimum” is false; the correct conclusion is

$\inf A = \frac{1}{2}, \quad A \text{ has no minimum.}$

Conclusion: This proof is completely acceptable!

□

Problem 4: Let $f(x) \geq 0$, $g(x) > 0$ be continuous on $[a, b]$. Prove that

$$\lim_{n \rightarrow \infty} \left(\int_a^b (f(x))^n g(x) dx \right)^{1/n} = \max_{a \leq x \leq b} f(x).$$

Proof from AI (DeepSeek):

Proof. Let

$$M = \max_{a \leq x \leq b} f(x).$$

Since f is continuous on the compact interval $[a, b]$, this maximum is attained at some point $x_0 \in [a, b]$, so $f(x_0) = M$.

Let

$$I_n = \int_a^b (f(x))^n g(x) dx.$$

We need to show

$$\lim_{n \rightarrow \infty} I_n^{1/n} = M.$$

(You must give a deep thought about why the maximum point of $f(x)$ matters here!)

Upper bound

Since $0 \leq f(x) \leq M$ for all $x \in [a, b]$, we have

$$(f(x))^n \leq M^n.$$

Therefore,

$$I_n = \int_a^b (f(x))^n g(x) dx \leq M^n \int_a^b g(x) dx.$$

Let

$$C = \int_a^b g(x) dx > 0.$$

Then

$$I_n^{1/n} \leq M C^{1/n}.$$

Since $C^{1/n} \rightarrow 1$ as $n \rightarrow \infty$, we get

$$\limsup_{n \rightarrow \infty} I_n^{1/n} \leq M.$$

Lower bound

Because f is continuous and $f(x_0) = M$, for every $\varepsilon > 0$, there exists a $\delta > 0$ such that

$$f(x) \geq M - \varepsilon$$

for all $x \in [a, b]$ with $|x - x_0| < \delta$.

Let $[c, d] \subset [a, b]$ be a subinterval around x_0 of positive length on which

$$f(x) \geq M - \varepsilon.$$

Since $g(x) > 0$ and continuous on $[a, b]$, it has a positive minimum on $[c, d]$, say

$$m = \min_{x \in [c, d]} g(x) > 0.$$

Then

$$I_n \geq \int_c^d (f(x))^n g(x) dx \geq (M - \varepsilon)^n \int_c^d g(x) dx \geq (M - \varepsilon)^n m(d - c).$$

Thus

$$I_n^{1/n} \geq (M - \varepsilon)(m(d - c))^{1/n}.$$

Since

$$(m(d - c))^{1/n} \rightarrow 1$$

as $n \rightarrow \infty$, we obtain

$$\liminf_{n \rightarrow \infty} I_n^{1/n} \geq M - \varepsilon.$$

Because $\varepsilon > 0$ is arbitrary,

$$\liminf_{n \rightarrow \infty} I_n^{1/n} \geq M.$$

Conclusion

Combining the upper and lower bounds gives

$$M \leq \liminf_{n \rightarrow \infty} I_n^{1/n} \leq \limsup_{n \rightarrow \infty} I_n^{1/n} \leq M.$$

Therefore,

$$\lim_{n \rightarrow \infty} I_n^{1/n} = M.$$

That is,

$$\lim_{n \rightarrow \infty} \left(\int_a^b (f(x))^n g(x) dx \right)^{1/n} = \max_{a \leq x \leq b} f(x).$$

This completes the proof.

(**Conclusion:** This proof is completely acceptable! Most importantly, you must realize that the maximum point plays a pivotal role in this type of asymptotic integral.) $\square$

Problem 5: Let $f(x)$ be integrable on $[a, b]$, and let $g(x) \geq 0$ be periodic with period $T > 0$ and integrable on $[0, T]$. Prove that

$$\lim_{n \rightarrow \infty} \int_a^b f(x) g(nx) dx = \frac{1}{T} \int_0^T g(x) dx \int_a^b f(x) dx.$$

Proof from AI (DeepSeek):

Proof. Let

$$\bar{g} = \frac{1}{T} \int_0^T g(x) dx.$$

We want to show

$$\lim_{n \rightarrow \infty} \int_a^b f(x) g(nx) dx = \bar{g} \int_a^b f(x) dx.$$

Define

$$h(x) = g(x) - \bar{g}.$$

Then h is also T -periodic and integrable on $[0, T]$, and

$$\int_0^T h(x) dx = 0.$$

Thus it suffices to prove

$$\lim_{n \rightarrow \infty} \int_a^b f(x) h(nx) dx = 0.$$

Indeed, once this is shown,

$$\int_a^b f(x) g(nx) dx = \bar{g} \int_a^b f(x) dx + \int_a^b f(x) h(nx) dx \rightarrow \bar{g} \int_a^b f(x) dx.$$

Step 1: Step functions

First suppose f is a step function. By linearity, it is enough to consider the case when f is the characteristic function of an interval $[\alpha, \beta] \subset [a, b]$. Then

$$\int_a^b f(x)h(nx) dx = \int_\alpha^\beta h(nx) dx.$$

Make the substitution $y = nx$. Then

$$\int_\alpha^\beta h(nx) dx = \frac{1}{n} \int_{n\alpha}^{n\beta} h(y) dy.$$

Since h is T -periodic and has zero mean over one period, its integral over any interval is bounded by a constant depending only on h . More precisely,

$$\left| \int_{n\alpha}^{n\beta} h(y) dy \right| \leq 2T \sup_{[0,T]} |h|$$

for all n . (This step is actually right, because $[n\alpha, n\beta]$ may be very long, and it contains many periods, on which the integrals are 0. In the end, there is only one part I remained, the length of which is less than a period T , and

$$\left| \int_{n\alpha}^{n\beta} h(y) dy \right| = \left| \int_I h(y) dy \right| \leq T \sup_{[0,T]} |h|.$$

)

Therefore,

$$\left| \int_\alpha^\beta h(nx) dx \right| \leq \frac{2T \sup_{[0,T]} |h|}{n} \rightarrow 0.$$

Hence the result holds for step functions.

Step 2: Extension to integrable functions

Let $\varepsilon > 0$. Since step functions are dense in $L^1[a, b]$, there exists a step function $s(x)$ such that

$$\int_a^b |f(x) - s(x)| dx < \varepsilon.$$

(Why is this true? how can we construct this approximation? This can be done by uniform continuity! because $f(x)$ is continuous, and therefore uniformly continuous

on $[a, b]$. For any $\epsilon > 0$, there exists $\delta > 0$ such that $\sup_{|x-x'|<\delta} |f(x) - f(x')| < \epsilon/(b-a)$. Therefore, we can uniformly decompose interval $[a, b]$ into $n = \lceil \frac{b-a}{\delta} \rceil + 1$ equal parts. Then

$$f_n(x) = \sum_{i=1}^n f\left(a + \frac{b-a}{n}(i-1)\right) I_{\left[a + \frac{b-a}{n}(i-1), a + \frac{b-a}{n}i\right)}(x)$$

where

$$I_{[c,d]}(x) = \begin{cases} 1, & x \in [c, d] \\ 0, & x \notin [c, d] \end{cases}$$

and

$$\sup_{x \in [a,b]} |f(x) - f_n(x)| < \epsilon/(b-a).$$

Then

$$\int_a^b |f(x) - f_n(x)| dx \leq \epsilon$$

)

Now write

$$\int_a^b f(x)h(nx) dx = \int_a^b (f(x) - s(x))h(nx) dx + \int_a^b s(x)h(nx) dx.$$

The second term tends to 0 by Step 1. For the first term,

$$\left| \int_a^b (f(x) - s(x))h(nx) dx \right| \leq \int_a^b |f(x) - s(x)| |h(nx)| dx \leq \sup |h| \int_a^b |f(x) - s(x)| dx \leq \sup |h| \epsilon.$$

Since $\epsilon > 0$ is arbitrary, it follows that

$$\lim_{n \rightarrow \infty} \int_a^b f(x)h(nx) dx = 0.$$

Conclusion

Thus

$$\lim_{n \rightarrow \infty} \int_a^b f(x)g(nx) dx = \bar{g} \int_a^b f(x) dx = \frac{1}{T} \int_0^T g(x) dx \int_a^b f(x) dx.$$

This completes the proof.

(This proof is acceptable and conventional. We should always start with simple functions such as $I_{[a,b]}(x)$, then we generalize to more general cases. But you may also use a different method to do it. For instance,

$$\begin{aligned}
\int_a^b f(x)g(nx) dx &= \sum_{i=0}^m \int_{a+i\frac{T}{n}}^{a+(i+1)\frac{T}{n}} f(x)h(nx)dx, \quad a + \frac{m+1}{n} = b, \\
&\approx \sum_{i=0}^m f(a + i\frac{T}{n}) \int_{a+i\frac{T}{n}}^{a+(i+1)\frac{T}{n}} h(nx)dx, \\
&= \sum_{i=0}^m \frac{1}{n} f(a + i\frac{T}{n}) \int_{na+iT}^{na+(i+1)T} h(y)dy, \quad nx = y \\
&= \frac{1}{T} \sum_{i=0}^m \frac{T}{n} f(a + i\frac{T}{n}) \int_0^T h(iT + u)du, \quad u = y - na \\
&= \frac{1}{T} \int_0^T h(u)du \sum_{i=0}^m \frac{T}{n} f(a + i\frac{T}{n}), \\
&\approx \frac{1}{T} \int_0^T h(u)du \int_a^b f(x)dx
\end{aligned}$$

)

□

Problem 6: Let $f(x)$ be twice continuously differentiable on $[0, 1]$ with $f(0) = f(1) = 0$ and $f(x) \neq 0$ for $x \in (0, 1)$. Prove that

$$\int_0^1 \left| \frac{f''(x)}{f(x)} \right| dx \geq 4.$$

Proof from AI (DeepSeek):

Proof. Let $f \in C^2[0, 1]$, $f(0) = f(1) = 0$, and $f(x) \neq 0$ for $x \in (0, 1)$. We want to prove

$$\int_0^1 \left| \frac{f''(x)}{f(x)} \right| dx \geq 4.$$

Since $f(0) = f(1) = 0$ and $f(x) \neq 0$ for $x \in (0, 1)$, by continuity f has a constant sign on $(0, 1)$. Without loss of generality, assume

$$f(x) > 0 \quad \text{for all } x \in (0, 1).$$

If $f(x) < 0$, we can replace f by $-f$, and the expression remains unchanged.

Because f is continuous on $[0, 1]$, it attains its maximum at some point $x_0 \in (0, 1)$.
Let

$$M = f(x_0) = \max_{[0,1]} f(x) > 0.$$

Step 1: Apply the mean value theorem on $[0, x_0]$

Since $f(0) = 0$ and $f(x_0) = M$, by the mean value theorem there exists $c_1 \in (0, x_0)$ such that

$$f'(c_1) = \frac{f(x_0) - f(0)}{x_0 - 0} = \frac{M}{x_0}.$$

Step 2: Apply the mean value theorem on $[x_0, 1]$

Similarly, since $f(x_0) = M$ and $f(1) = 0$, there exists $c_2 \in (x_0, 1)$ such that

$$f'(c_2) = \frac{f(1) - f(x_0)}{1 - x_0} = -\frac{M}{1 - x_0}.$$

Step 3: Estimate the integral

Since $f(x) > 0$, we have

$$\int_0^1 \left| \frac{f''(x)}{f(x)} \right| dx \geq \int_0^1 \frac{|f''(x)|}{M} dx = \frac{1}{M} \int_0^1 |f''(x)| dx.$$

Now,

$$\int_0^1 |f''(x)| dx \geq \left| \int_{c_1}^{c_2} f''(x) dx \right| = |f'(c_2) - f'(c_1)|.$$

(Because $f''(x)$ changes sign in $[c_1, c_2] \subset [0, 1]$, then

$$\int_0^1 |f''(x)| dx \geq \int_{c_1}^{c_2} |f''(x)| dx \geq \left| \int_{c_1}^{c_2} f''(x) dx \right|$$

)

Using the values of $f'(c_1)$ and $f'(c_2)$,

$$|f'(c_2) - f'(c_1)| = \left| -\frac{M}{1 - x_0} - \frac{M}{x_0} \right| = M \left(\frac{1}{1 - x_0} + \frac{1}{x_0} \right).$$

Therefore,

$$\int_0^1 \left| \frac{f''(x)}{f(x)} \right| dx \geq \frac{1}{M} \cdot M \left(\frac{1}{1 - x_0} + \frac{1}{x_0} \right) = \frac{1}{1 - x_0} + \frac{1}{x_0}.$$

Step 4: Minimize the lower bound

For $0 < x_0 < 1$,

$$\frac{1}{1-x_0} + \frac{1}{x_0} = \frac{1}{x_0(1-x_0)}.$$

The function $x_0(1-x_0)$ has maximum $\frac{1}{4}$ at $x_0 = \frac{1}{2}$. Hence

$$\frac{1}{x_0(1-x_0)} \geq 4.$$

Thus

$$\int_0^1 \left| \frac{f''(x)}{f(x)} \right| dx \geq 4.$$

This completes the proof.

(This proof is completely acceptable!)

□

Problem 7: Given that $\int_0^{+\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$, prove that

$$\int_0^{+\infty} e^{-a^2x^2 - \frac{b^2}{x^2}} dx = \frac{\sqrt{\pi}}{2a} e^{-2ab}.$$

Proof from AI (DeepSeek):

Proof. We are given

$$\int_0^{\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$$

and want to prove

$$\int_0^{\infty} e^{-a^2x^2 - \frac{b^2}{x^2}} dx = \frac{\sqrt{\pi}}{2a} e^{-2ab}.$$

Let

$$I(a, b) = \int_0^{\infty} e^{-a^2x^2 - \frac{b^2}{x^2}} dx.$$

We assume $a > 0$ and $b > 0$.

Make the substitution

$$u = ax.$$

Then $dx = \frac{du}{a}$, and

$$a^2x^2 = u^2, \quad \frac{b^2}{x^2} = \frac{a^2b^2}{u^2}.$$

Thus

$$I(a, b) = \frac{1}{a} \int_0^\infty e^{-u^2 - \frac{a^2 b^2}{u^2}} du.$$

Let

$$c = ab > 0.$$

Then

$$I(a, b) = \frac{1}{a} J(c),$$

where

$$J(c) = \int_0^\infty e^{-u^2 - \frac{c^2}{u^2}} du.$$

So it suffices to prove

$$J(c) = \frac{\sqrt{\pi}}{2} e^{-2c}.$$

Step 1: Differentiate $J(c)$

Differentiating under the integral sign,

$$J'(c) = \int_0^\infty e^{-u^2 - \frac{c^2}{u^2}} \left(-\frac{2c}{u^2} \right) du.$$

Make the substitution

$$v = \frac{c}{u}.$$

Then

$$u = \frac{c}{v}, \quad du = -\frac{c}{v^2} dv.$$

When $u \rightarrow 0^+$, $v \rightarrow +\infty$, and when $u \rightarrow +\infty$, $v \rightarrow 0^+$. Thus

$$J'(c) = \int_\infty^0 e^{-\frac{c^2}{v^2} - v^2} \left(-\frac{2c}{c^2/v^2} \right) \left(-\frac{c}{v^2} \right) dv.$$

Simplify the integrand:

$$-\frac{2c}{u^2} = -\frac{2c}{c^2/v^2} = -\frac{2v^2}{c}.$$

Also $du = -\frac{c}{v^2} dv$. Therefore,

$$J'(c) = \int_\infty^0 e^{-v^2 - \frac{c^2}{v^2}} \left(-\frac{2v^2}{c} \right) \left(-\frac{c}{v^2} \right) dv.$$

The product

$$\left(-\frac{2v^2}{c}\right)\left(-\frac{c}{v^2}\right) = 2.$$

Thus

$$J'(c) = \int_{-\infty}^0 2e^{-v^2 - \frac{c^2}{v^2}} dv = -2 \int_0^{\infty} e^{-v^2 - \frac{c^2}{v^2}} dv.$$

But the last integral is exactly $J(c)$. Hence

$$J'(c) = -2J(c).$$

Step 2: Solve the differential equation

We have

$$J'(c) = -2J(c).$$

The solution is

$$J(c) = J(0)e^{-2c}.$$

Now

$$J(0) = \int_0^{\infty} e^{-u^2} du = \frac{\sqrt{\pi}}{2}.$$

Therefore,

$$J(c) = \frac{\sqrt{\pi}}{2} e^{-2c}.$$

Step 3: Conclude

Recall $c = ab$. Then

$$I(a, b) = \frac{1}{a} J(ab) = \frac{1}{a} \cdot \frac{\sqrt{\pi}}{2} e^{-2ab} = \frac{\sqrt{\pi}}{2a} e^{-2ab}.$$

Thus

$$\boxed{\int_0^{\infty} e^{-a^2 x^2 - \frac{b^2}{x^2}} dx = \frac{\sqrt{\pi}}{2a} e^{-2ab}}.$$

(This proof is completely acceptable!)

□

5 Topic 4: Special functions and function space

We also have 3 hours lecture to cover all concepts. We will use 1 hour tutorial to talk two examples.

Implementation Demo:

3 hour lecture:

- Introduce the concept of convergence of a sequence of functions, and the uniform convergence of a sequence of functions.
- Introduce the criteria of uniform convergence of functions.
- Introduce the concept of a series of functions, such as power series and general sequences of functions. Approximation of polynomials.
- Introduce special series, such as geometric series, exponential series and Newton's Binomial theorem. We also talk about the manipulation of series (algebraic operations, differentiation and integration).
- Introduce Gamma integral and Beta integrals.

1 hour Tutorial:

Problem 1: Use the known formula

$$\int_{-1}^1 \frac{dx}{(\alpha - x)\sqrt{1 - x^2}} = \frac{\pi}{\sqrt{\alpha^2 - 1}}, \quad \alpha > 1,$$

to prove

$$\int_{-1}^1 \frac{x^{2n}}{\sqrt{1 - x^2}} dx = \frac{(2n - 1)!!}{(2n)!!} \pi, \quad n = 1, 2, \dots$$

Solution from AI (DeepSeek):

Proof. **Step 1: Expand the known formula**

Treat the left-hand side as a function of α . For $\alpha > 1$,

$$\frac{1}{\alpha - x} = \frac{1}{\alpha} \cdot \frac{1}{1 - \frac{x}{\alpha}} = \sum_{k=0}^{\infty} \frac{x^k}{\alpha^{k+1}}.$$

This series converges absolutely for $|x| \leq 1$ and $\alpha > 1$, so we can integrate term by term:

$$\int_{-1}^1 \frac{dx}{(\alpha - x)\sqrt{1 - x^2}} = \sum_{k=0}^{\infty} \frac{1}{\alpha^{k+1}} \int_{-1}^1 \frac{x^k}{\sqrt{1 - x^2}} dx.$$

Since $\frac{x^k}{\sqrt{1-x^2}}$ is an odd function on $[-1, 1]$ when k is odd, its integral is zero. Therefore only even powers remain:

$$\int_{-1}^1 \frac{dx}{(\alpha - x)\sqrt{1-x^2}} = \sum_{n=0}^{\infty} \frac{1}{\alpha^{2n+1}} \int_{-1}^1 \frac{x^{2n}}{\sqrt{1-x^2}} dx.$$

Step 2: Expand the right-hand side

For $\alpha > 1$,

$$\frac{\pi}{\sqrt{\alpha^2 - 1}} = \frac{\pi}{\alpha} \left(1 - \frac{1}{\alpha^2}\right)^{-1/2}.$$

Using the binomial series

$$(1 - t)^{-1/2} = \sum_{n=0}^{\infty} \binom{-\frac{1}{2}}{n} (-t)^n = \sum_{n=0}^{\infty} \frac{(2n-1)!!}{(2n)!!} t^n,$$

with $t = \frac{1}{\alpha^2}$, we obtain

$$\frac{\pi}{\sqrt{\alpha^2 - 1}} = \frac{\pi}{\alpha} \sum_{n=0}^{\infty} \frac{(2n-1)!!}{(2n)!!} \frac{1}{\alpha^{2n}} = \sum_{n=0}^{\infty} \frac{(2n-1)!!}{(2n)!!} \frac{\pi}{\alpha^{2n+1}}.$$

Step 3: Compare coefficients

Both sides are now power series in $\frac{1}{\alpha}$:

$$\sum_{n=0}^{\infty} \frac{1}{\alpha^{2n+1}} \int_{-1}^1 \frac{x^{2n}}{\sqrt{1-x^2}} dx = \sum_{n=0}^{\infty} \frac{(2n-1)!!}{(2n)!!} \frac{\pi}{\alpha^{2n+1}}.$$

Equating coefficients of $\frac{1}{\alpha^{2n+1}}$, we get

$$\int_{-1}^1 \frac{x^{2n}}{\sqrt{1-x^2}} dx = \frac{(2n-1)!!}{(2n)!!} \pi.$$

This proves the desired formula.

(This solution is completely acceptable!)

□

Problem 2: Let the sequence $\{y_n(x)\}$ satisfy the equation

$$\frac{d}{dx} \left[p(x) \frac{dy_n}{dx} \right] + \lambda_n y_n = 0 \quad (\forall x \in [a, b]) \quad (n = 1, 2, \dots) \quad (1)$$

(where $\lambda_n \neq \lambda_m$ when $n \neq m$) and the boundary conditions

$$y_n(a) = y_n(b) = 0.$$

Prove that $\{y_n(x)\}$ is an orthogonal system on $[a, b]$.

Solution from AI (DeepSeek):

Proof. We are given the sequence $\{y_n(x)\}$ satisfying the Sturm–Liouville equation

$$\frac{d}{dx} \left[p(x) \frac{dy_n}{dx} \right] + \lambda_n y_n = 0, \quad x \in [a, b],$$

with boundary conditions

$$y_n(a) = y_n(b) = 0,$$

and $\lambda_n \neq \lambda_m$ for $n \neq m$. We want to prove that $\{y_n(x)\}$ is an orthogonal system on $[a, b]$, that is,

$$\int_a^b y_n(x) y_m(x) dx = 0$$

for $n \neq m$.

Take two distinct indices n and m . The corresponding equations are

$$\frac{d}{dx} [p(x) y_n'(x)] + \lambda_n y_n(x) = 0,$$

$$\frac{d}{dx} [p(x) y_m'(x)] + \lambda_m y_m(x) = 0.$$

Multiply the first equation by y_m , the second by y_n , and subtract:

$$y_m \frac{d}{dx} [p y_n'] - y_n \frac{d}{dx} [p y_m'] + (\lambda_n - \lambda_m) y_n y_m = 0.$$

Thus

$$(\lambda_n - \lambda_m) y_n y_m = y_n \frac{d}{dx} [p y_m'] - y_m \frac{d}{dx} [p y_n'].$$

Now integrate both sides from a to b :

$$(\lambda_n - \lambda_m) \int_a^b y_n y_m dx = \int_a^b \left(y_n \frac{d}{dx} [p y_m'] - y_m \frac{d}{dx} [p y_n'] \right) dx.$$

The right-hand side is the integral of a derivative of a product. Indeed,

$$\frac{d}{dx} (py_n y'_m - py_m y'_n) = y_n \frac{d}{dx} [py'_m] + py'_n y'_m - y_m \frac{d}{dx} [py'_n] - py'_m y'_n.$$

The two middle terms cancel, so

$$\frac{d}{dx} (py_n y'_m - py_m y'_n) = y_n \frac{d}{dx} [py'_m] - y_m \frac{d}{dx} [py'_n].$$

Therefore,

$$(\lambda_n - \lambda_m) \int_a^b y_n y_m dx = [py_n y'_m - py_m y'_n]_a^b.$$

Using the boundary conditions $y_n(a) = y_n(b) = 0$ and $y_m(a) = y_m(b) = 0$, we get

$$[py_n y'_m - py_m y'_n]_a^b = 0.$$

Hence

$$(\lambda_n - \lambda_m) \int_a^b y_n y_m dx = 0.$$

Since $\lambda_n \neq \lambda_m$, we have $\lambda_n - \lambda_m \neq 0$. Therefore,

$$\int_a^b y_n y_m dx = 0.$$

Thus $\{y_n(x)\}$ is an orthogonal system on $[a, b]$.

(This proof is completely acceptable!)

□

6 Topic 5: Calculus of multivariate function

We are supposed to complete this topic in three weeks. So we have 9 hours lecture and 3 hours tutorial.

Implementation Demo:

9 hours lecture:

- We should cover basic topology of space $\mathbb{R}^n$.
- Then we discuss the definition of limit of multivariate function and its continuity.

- We will also talk about the differentiability, mean value theorem and extreme value theorems.
- Then we continue to cover the integral of scalar multivariate function, including substitution and integration by parts.
- In the end, we also talk about the integral of vector-valued multivariate function, it is called integral of differential form.

3 hours tutorial:

Problem 1: Let $f(x)$ be continuous on $\mathbf{R}^n$ with $f(x) > 0$ when $x \neq 0$ (the origin in $\mathbf{R}^n$), and suppose that for all $x \in \mathbf{R}^n$ and all $c > 0$, $f(cx) = cf(x)$. Prove that there exist $a, b > 0$ such that

$$a|x| \leq f(x) \leq b|x| \quad (\forall x \in \mathbf{R}^n).$$

Solution from AI (DeepSeek):

Proof. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be continuous and satisfy

$$f(x) > 0 \quad \text{for } x \neq 0,$$

and

$$f(cx) = cf(x)$$

for all $x \in \mathbb{R}^n$ and all $c > 0$. We want to prove that there exist constants $a, b > 0$ such that

$$a|x| \leq f(x) \leq b|x|$$

for all $x \in \mathbb{R}^n$.

Let $S^{n-1} = \{x \in \mathbb{R}^n : |x| = 1\}$ be the unit sphere in $\mathbb{R}^n$. This set is compact because it is closed and bounded.

Since f is continuous and positive on S^{n-1} , the restriction $f|_{S^{n-1}}$ attains a positive minimum and a positive maximum. Thus there exist constants $a, b > 0$ such that

$$a \leq f(u) \leq b$$

for all $u \in S^{n-1}$.

Now take any $x \in \mathbb{R}^n$.

If $x = 0$, then by homogeneity with $c = 2$,

$$f(0) = f(2 \cdot 0) = 2f(0),$$

so $f(0) = 0$. Hence

$$a|0| = 0 \leq f(0) = 0 \leq b|0| = 0.$$

If $x \neq 0$, let

$$u = \frac{x}{|x|}.$$

Then $u \in S^{n-1}$ and

$$x = |x|u.$$

Using homogeneity with $c = |x| > 0$, we get

$$f(x) = f(|x|u) = |x|f(u).$$

Since $a \leq f(u) \leq b$, multiplying by $|x| > 0$ gives

$$a|x| \leq |x|f(u) \leq b|x|.$$

Therefore,

$$a|x| \leq f(x) \leq b|x|.$$

Thus the inequality holds for all $x \in \mathbb{R}^n$.

This completes the proof.

(This proof is completely acceptable!)

□

Problem 2: Transform the equation $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0$ into polar coordinates.

Solution from AI (DeepSeek):

Proof. : We start with Laplace's equation in Cartesian coordinates:

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0.$$

We want to express this equation in polar coordinates (r, θ) , where

$$x = r \cos \theta, \quad y = r \sin \theta.$$

Step 1: Chain rule relations

We have

$$r = \sqrt{x^2 + y^2}, \quad \theta = \arctan \frac{y}{x}.$$

The first-order partial derivatives transform as

$$\frac{\partial}{\partial x} = \frac{\partial r}{\partial x} \frac{\partial}{\partial r} + \frac{\partial \theta}{\partial x} \frac{\partial}{\partial \theta},$$

$$\frac{\partial}{\partial y} = \frac{\partial r}{\partial y} \frac{\partial}{\partial r} + \frac{\partial \theta}{\partial y} \frac{\partial}{\partial \theta}.$$

Now

$$\frac{\partial r}{\partial x} = \frac{x}{r} = \cos \theta, \quad \frac{\partial r}{\partial y} = \frac{y}{r} = \sin \theta,$$

and

$$\begin{aligned} \frac{\partial \theta}{\partial x} &= -\frac{y}{x^2 + y^2} = -\frac{\sin \theta}{r}, \\ \frac{\partial \theta}{\partial y} &= \frac{x}{x^2 + y^2} = \frac{\cos \theta}{r}. \end{aligned}$$

Thus

$$\begin{aligned} \frac{\partial}{\partial x} &= \cos \theta \frac{\partial}{\partial r} - \frac{\sin \theta}{r} \frac{\partial}{\partial \theta}, \\ \frac{\partial}{\partial y} &= \sin \theta \frac{\partial}{\partial r} + \frac{\cos \theta}{r} \frac{\partial}{\partial \theta}. \end{aligned}$$

Step 2: Second-order derivatives

Apply the operator for $\frac{\partial}{\partial x}$ twice:

$$\frac{\partial^2}{\partial x^2} = \left(\cos \theta \frac{\partial}{\partial r} - \frac{\sin \theta}{r} \frac{\partial}{\partial \theta} \right) \left(\cos \theta \frac{\partial}{\partial r} - \frac{\sin \theta}{r} \frac{\partial}{\partial \theta} \right).$$

Expanding carefully,

$$\frac{\partial^2}{\partial x^2} = \cos^2 \theta \frac{\partial^2}{\partial r^2} - \frac{2 \sin \theta \cos \theta}{r} \frac{\partial^2}{\partial r \partial \theta} + \frac{\sin^2 \theta}{r^2} \frac{\partial^2}{\partial \theta^2} + \frac{\sin^2 \theta}{r} \frac{\partial}{\partial r} + \frac{2 \sin \theta \cos \theta}{r^2} \frac{\partial}{\partial \theta}.$$

Similarly,

$$\frac{\partial^2}{\partial y^2} = \sin^2 \theta \frac{\partial^2}{\partial r^2} + \frac{2 \sin \theta \cos \theta}{r} \frac{\partial^2}{\partial r \partial \theta} + \frac{\cos^2 \theta}{r^2} \frac{\partial^2}{\partial \theta^2} + \frac{\cos^2 \theta}{r} \frac{\partial}{\partial r} - \frac{2 \sin \theta \cos \theta}{r^2} \frac{\partial}{\partial \theta}.$$

Step 3: Add the two second derivatives

Adding the two expressions, the mixed derivative terms cancel, and we get

$$\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} = \frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial}{\partial r} + \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2}.$$

Therefore, Laplace's equation in polar coordinates is

$$\frac{\partial^2 z}{\partial r^2} + \frac{1}{r} \frac{\partial z}{\partial r} + \frac{1}{r^2} \frac{\partial^2 z}{\partial \theta^2} = 0.$$

Thus,

$$\boxed{\frac{\partial^2 z}{\partial r^2} + \frac{1}{r} \frac{\partial z}{\partial r} + \frac{1}{r^2} \frac{\partial^2 z}{\partial \theta^2} = 0}$$

(This proof is completely acceptable!)

□

Problem 3: Let $F(x, y, z)$ have continuous first-order partial derivatives $\frac{\partial F}{\partial x}, \frac{\partial F}{\partial y}, \frac{\partial F}{\partial z}$ on $\mathbf{R}^3$ and satisfy the inequality

$$y \frac{\partial F}{\partial x} - x \frac{\partial F}{\partial y} + \frac{\partial F}{\partial z} \geq \alpha > 0, \quad \forall (x, y, z) \in \mathbf{R}^3,$$

where α is a constant. Prove that as the point (x, y, z) goes to infinity along the curve

$$\Gamma: \quad x = -\cos t, \quad y = \sin t, \quad z = t, \quad t \geq 0,$$

we have $F(x, y, z) \rightarrow +\infty$.

Solution from AI (DeepSeek):

Proof. : Let

$$\varphi(t) = F(-\cos t, \sin t, t).$$

Since F has continuous first-order partial derivatives, φ is differentiable for $t \geq 0$. By the chain rule,

$$\varphi'(t) = \frac{\partial F}{\partial x} \frac{dx}{dt} + \frac{\partial F}{\partial y} \frac{dy}{dt} + \frac{\partial F}{\partial z} \frac{dz}{dt}.$$

Now

$$\frac{dx}{dt} = \sin t, \quad \frac{dy}{dt} = \cos t, \quad \frac{dz}{dt} = 1.$$

Thus

$$\varphi'(t) = \sin t \frac{\partial F}{\partial x} + \cos t \frac{\partial F}{\partial y} + \frac{\partial F}{\partial z}.$$

But on Γ , we have

$$y = \sin t, \quad -x = \cos t.$$

Therefore,

$$\varphi'(t) = y \frac{\partial F}{\partial x} - x \frac{\partial F}{\partial y} + \frac{\partial F}{\partial z}.$$

Using the given inequality,

$$\varphi'(t) \geq \alpha > 0$$

for all $t \geq 0$.

Hence

$$\varphi(t) - \varphi(0) = \int_0^t \varphi'(s) ds \geq \int_0^t \alpha ds = \alpha t.$$

Therefore,

$$\varphi(t) \geq \varphi(0) + \alpha t.$$

Since $\alpha > 0$, we have

$$\alpha t \rightarrow +\infty$$

as $t \rightarrow +\infty$. Hence

$$\varphi(t) \rightarrow +\infty.$$

Thus

$$F(-\cos t, \sin t, t) \rightarrow +\infty$$

as $t \rightarrow +\infty$.

This completes the proof.

(This proof is completely acceptable!)

□

Problem 4: Find the extrema of the function $f(x, y, z) = x^4 + y^4 + z^4$ under the condition $xyz = 1$. Are these extrema maxima or minima? Why?

Solution from AI (DeepSeek):

Solution: We want to find the extrema of

$$f(x, y, z) = x^4 + y^4 + z^4$$

subject to the constraint

$$xyz = 1.$$

Step 1: Lagrange multipliers

Let

$$g(x, y, z) = xyz - 1 = 0.$$

Then

$$\nabla f = \lambda \nabla g.$$

Compute the gradients:

$$\nabla f = (4x^3, 4y^3, 4z^3),$$

$$\nabla g = (yz, xz, xy).$$

Thus we have the system

$$4x^3 = \lambda yz,$$

$$4y^3 = \lambda xz,$$

$$4z^3 = \lambda xy,$$

together with

$$xyz = 1.$$

Step 2: Solve the system

Multiply the first equation by x , the second by y , and the third by z :

$$4x^4 = \lambda xyz,$$

$$4y^4 = \lambda xyz,$$

$$4z^4 = \lambda xyz.$$

Since $xyz = 1$, we get

$$4x^4 = 4y^4 = 4z^4 = \lambda.$$

Therefore,

$$x^4 = y^4 = z^4.$$

Since $x^4, y^4, z^4 \geq 0$, this implies

$$|x| = |y| = |z|.$$

Step 3: Use the constraint

From $xyz = 1$, the product is positive. Therefore either all three variables are positive, or exactly two are negative and one is positive.

Case 1: $x, y, z > 0$ Then $x = y = z$. The constraint gives

$$x^3 = 1,$$

so

$$x = y = z = 1.$$

The value of f is

$$f(1, 1, 1) = 1 + 1 + 1 = 3.$$

Case 2: Exactly two negative, one positive Since $|x| = |y| = |z|$, let the positive one be $a > 0$ and the two negative ones be $-a$. Then

$$xyz = a(-a)(-a) = a^3 = 1,$$

so $a = 1$. Thus the points are permutations of

$$(1, -1, -1).$$

At these points,

$$f(1, -1, -1) = 1 + 1 + 1 = 3.$$

Thus all candidates give the same value 3.

Step 4: Are they minima or maxima?

We have found that all points satisfying the constraint and the Lagrange multiplier equations give $f = 3$.

Now consider the behavior of f on the constraint set. Since $xyz = 1$, at least one variable must be nonzero. The function

$$f(x, y, z) = x^4 + y^4 + z^4$$

is coercive on the constraint set in the sense that if any variable tends to infinity, then f tends to infinity. More precisely, if $|x| \rightarrow \infty$, then because $xyz = 1$, at least one of y, z may adjust, but the sum of fourth powers will still tend to infinity. Thus f cannot have a maximum; it grows without bound.

Therefore the value 3 is the global minimum, and there is no maximum.

Conclusion

The extrema under the constraint $xyz = 1$ are

$$(1, 1, 1)$$

and all permutations of

$$(1, -1, -1).$$

At all these points,

$$f = 3.$$

These are global minima. There is no global maximum because f tends to $+\infty$ along the constraint set.

(This proof is completely acceptable!)

Problem 5: Compute the integral

$$I = \iint_D \frac{(x+y) \ln(1 + \frac{y}{x})}{\sqrt{2-x-y}} dx dy,$$

where D is the triangular region bounded by $y = 0, y = x, x + y = 1$.

Solution from AI (DeepSeek):

Solution: We compute

$$I = \iint_D \frac{(x+y) \ln(1 + \frac{y}{x})}{\sqrt{2-x-y}} dx dy,$$

where D is the triangle bounded by

$$y = 0, \quad y = x, \quad x + y = 1.$$

Step 1: Change variables

Let

$$u = x + y, \quad v = \frac{y}{x}.$$

Then

$$x = \frac{u}{1+v}, \quad y = \frac{uv}{1+v}.$$

The boundaries become

$$v = 0, \quad v = 1, \quad u = 1,$$

so the region transforms to

$$0 \leq u \leq 1, \quad 0 \leq v \leq 1.$$

Step 2: Jacobian

We compute

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{1}{1+v} & -\frac{u}{(1+v)^2} \\ \frac{v}{1+v} & \frac{u}{(1+v)^2} \end{vmatrix} = \frac{u}{(1+v)^2}.$$

Thus

$$dx dy = \frac{u}{(1+v)^2} du dv.$$

Also,

$$x + y = u, \quad \ln \left(1 + \frac{y}{x} \right) = \ln(1 + v).$$

Therefore,

$$I = \int_0^1 \int_0^1 \frac{u^2 \ln(1 + v)}{(1 + v)^2 \sqrt{2 - u}} du dv.$$

Step 3: Separate the integrals

$$I = \left(\int_0^1 \frac{u^2}{\sqrt{2 - u}} du \right) \left(\int_0^1 \frac{\ln(1 + v)}{(1 + v)^2} dv \right).$$

Step 4: Compute the u -integral

Let

$$A = \int_0^1 \frac{u^2}{\sqrt{2 - u}} du.$$

Using $t = 2 - u$,

$$A = \int_1^2 \frac{(2 - t)^2}{\sqrt{t}} dt = \int_1^2 \left(4t^{-1/2} - 4t^{1/2} + t^{3/2} \right) dt.$$

Thus

$$A = \left[8t^{1/2} - \frac{8}{3}t^{3/2} + \frac{2}{5}t^{5/2} \right]_1^2 = \frac{64\sqrt{2} - 86}{15}.$$

Step 5: Compute the v -integral

Let

$$B = \int_0^1 \frac{\ln(1 + v)}{(1 + v)^2} dv.$$

With $w = 1 + v$,

$$B = \int_1^2 \frac{\ln w}{w^2} dw.$$

Integration by parts gives

$$\int \frac{\ln w}{w^2} dw = -\frac{\ln w}{w} - \frac{1}{w} + C.$$

Therefore,

$$B = \left[-\frac{\ln w}{w} - \frac{1}{w} \right]_1^2 = \frac{1 - \ln 2}{2}.$$

Step 6: Combine

$$I = A \cdot B = \frac{64\sqrt{2} - 86}{15} \cdot \frac{1 - \ln 2}{2} = \frac{(64\sqrt{2} - 86)(1 - \ln 2)}{30}.$$

Thus,

$$I = \frac{(64\sqrt{2} - 86)(1 - \ln 2)}{30}$$

(This proof is completely acceptable!)

Problem 6: Compute the improper integral

$$\iint_D x^{-\frac{3}{2}} e^{y-x} dx dy,$$

where $D : y \geq 0, x \geq y$.

Solution from AI (DeepSeek):

Solution: We compute the improper integral

$$I = \iint_D x^{-3/2} e^{y-x} dx dy,$$

where

$$D = \{(x, y) : y \geq 0, x \geq y\}.$$

Step 1: Rewrite the region

The region is described by

$$0 \leq y \leq x, \quad x \geq 0.$$

Thus

$$I = \int_0^\infty \int_0^x x^{-3/2} e^{y-x} dy dx.$$

Step 2: Integrate with respect to y

The inner integral is

$$\int_0^x x^{-3/2} e^{y-x} dy = x^{-3/2} e^{-x} \int_0^x e^y dy.$$

Since

$$\int_0^x e^y dy = e^x - 1,$$

we get

$$\int_0^x x^{-3/2} e^{y-x} dy = x^{-3/2} (1 - e^{-x}).$$

Thus

$$I = \int_0^\infty x^{-3/2} (1 - e^{-x}) dx.$$

Step 3: Evaluate the remaining integral

We need

$$I = \int_0^\infty \frac{1 - e^{-x}}{x^{3/2}} dx.$$

Integration by parts: let

$$u = 1 - e^{-x}, \quad dv = x^{-3/2} dx.$$

Then

$$du = e^{-x} dx, \quad v = -2x^{-1/2}.$$

Thus

$$I = \left[-2x^{-1/2} (1 - e^{-x}) \right]_0^\infty + 2 \int_0^\infty x^{-1/2} e^{-x} dx.$$

The boundary term vanishes because

$$\lim_{x \rightarrow \infty} -2x^{-1/2} (1 - e^{-x}) = 0$$

and

$$\lim_{x \rightarrow 0^+} -2x^{-1/2} (1 - e^{-x}) = 0.$$

Therefore,

$$I = 2 \int_0^\infty x^{-1/2} e^{-x} dx.$$

Step 4: Recognize the gamma function

$$\int_0^\infty x^{-1/2} e^{-x} dx = \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}.$$

Hence

$$I = 2\sqrt{\pi}.$$

Thus,

$$\boxed{2\sqrt{\pi}}$$

(This proof is completely acceptable!)

7 Topic 6: Lebesgue integral

We will only use one week to cover basic definitions of Lebesgue measure and its integral. But this will not be tested in the final. So it may be ignored in the end if our teaching time is tight. We will leave this open for discussion.